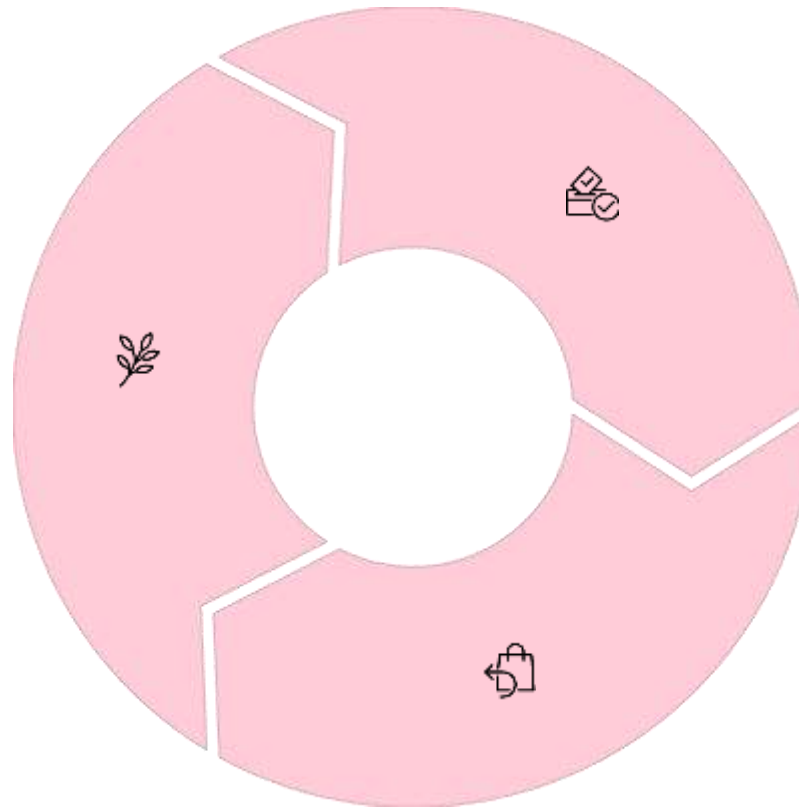


Branch Prediction

Modern processors use branch prediction to avoid pipeline stalls. The Branch Prediction Unit (BPU) predicts branch targets and outcomes before actual determination, enabling speculative execution that improves performance.

Branch Target Buffer

Stores computed target addresses of previously taken branch instructions for quick prediction.



Directional Predictor

Predicts whether branches will be taken or not taken based on historical patterns.

Return Stack Buffer

Predicts return addresses by maintaining a hardware stack of call instruction returns.

While improving performance, these prediction mechanisms create attack surfaces that leak sensitive control flow information.

PHT-Based Attacks

- Page History Table attacks exploit the directional branch predictor, creating collisions between victim and attacker process branches to infer branch directions and extract sensitive information.
- BranchScope
 - First attack exploiting PHT as an attack surface. Requires attacker and victim on same core, victim slowdown for high-resolution observation, and triggering many branches to activate vulnerable branches.
- Bluethunder
 - Evolutionary improvement creating collisions in 2-level predictors. Requires only single trigger of 2-level predictor versus many branches for bimodel predictor, achieving 52× faster attacks than BranchScope.

SgxPectre: Spectre Meets SGX

- SgxPectre extends the Spectre attack to compromise SGX enclaves by exploiting speculative execution channels. This attack manipulates the Branch Target Buffer to temporarily control enclave program control flow.
- The attack proceeds in five steps:
 - Poison the BTB to cause misprediction toward secret-leaking instructions
 - Increase probability of executing those instructions
 - Set registers for secret-leaking operations
 - Trigger enclave code execution
 - Extract secrets via cache channels using Flush+Reload
- Successfully demonstrated stealing seal keys and attestation keys from Intel-signed quoting enclaves.

RSB-Based Attacks

Return Stack Buffer attacks exploit the hardware stack used for predicting return addresses. Unlike BTB-based attacks, It pollutes the RSB to trigger mispredictions that leak sensitive information.

1

RSB Pollution

Attacker fills RSB with target addresses pointing to malicious payload gadgets containing secret-leaking instruction sequences.

2

Enclave Invocation

Attacker invokes specific enclave call to switch to trusted execution mode with polluted RSB state.

3

Unmatched Return

Enclave call contains unmatched return causing speculative execution at malicious target address.

4

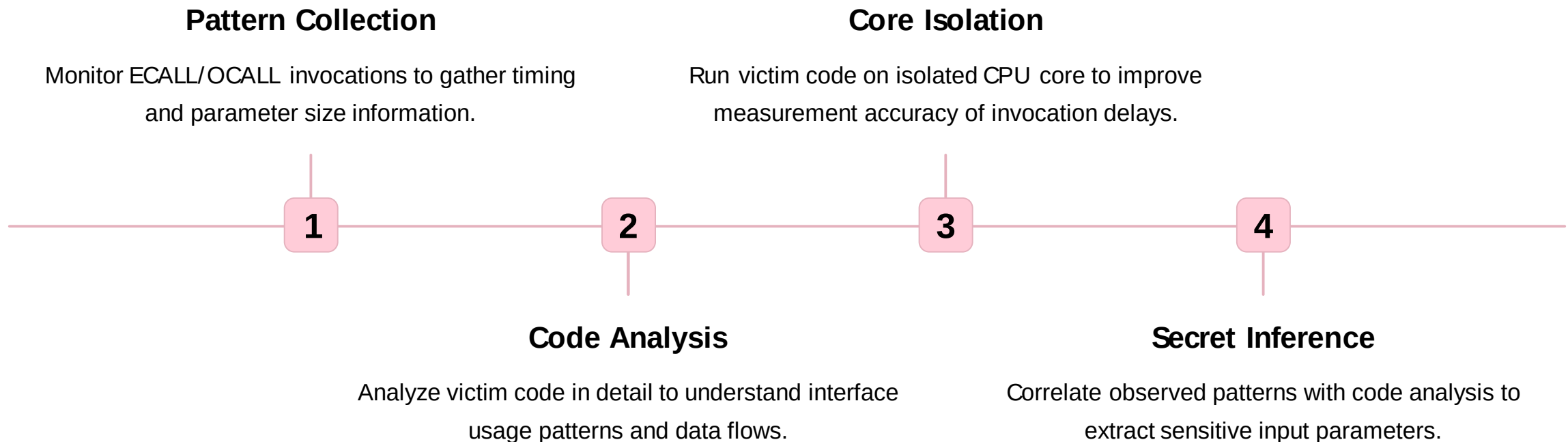
Secret Extraction

Attacker exploits cache channel to infer leaked information from speculatively executed malicious gadgets.

Attacks succeed even against fully patched machines, demonstrating persistent security risks.

Enclave Interface Attacks

Enclave interface attacks exploit observable patterns in ECALL/OCALL function invocations. Attackers collect information about input parameter sizes and function invocation delays to infer sensitive data.



Demonstrated attacks against SGX-assisted network function virtualization platforms prove effectiveness of interface pattern analysis.